Trade & Ahead financial consultancy firm

Unsupervised Learning

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Contents / Agenda

Executive Summary

- This is a small dataset that categorizes 340 companies according to the Global Industry Classification Standard. It is not clear if this is a deliberate selection of companies to attempt to meet a particular standard Market Index (such as the Dow Jones Market, Nasdaq, or S&P 500 indices).

- We applied two different clustering techniques to gain more insight into the company characteristics. The K-Means Clustering technique proved to be the more useful method for this dataset. We were able to distinguish four clusters and provide you with the following observations:

- **Cluster 0** contains mostly companies from the Energy sector (22 out of 27), and they stand out in Volatility, Price_Change, ROE, and P/E_Ratio. In general these attributes suggest high growth companies with potentially higher risk and reward. In this cluster, the Energy companies may have experienced extreme fluctuation in oil and gas prices.

- **Cluster 1** contains companies that stand out in categories Net_Income and Estimated_Shares_Outstanding. They are the least Volatile and subjective to Price_Change. This suggests these are likely large, well-established companies with a mature business model.

Executive Summary continued

- **Cluster 2** contains the vast majority of companies in the dataset: 277 out of 340. No other particular attribute stood out in the cluster analysis for this cluster.

- **Cluster 3** contains companies that stand out in Current_Price, Price_Change, Cash_Ratio, Net_Cash_Flow, P/B_Ratio. To a lesser extent: Earnings_Per_Share. This paints a picture of potentially financially strong, established companies with recent price momentum.

- Based on our analysis of these clusters, we recommend to pay attention to the companies listed in cluster 1 and 3, for adding stability and dividends in a customer’s diversified portfolio.

Business Problem Overview and Solution Approach

- A diversified portfolio is important to maintain when one is investing in the stock market, in order to maximise earnings under any market condition. A diversified portfolio tends to yield higher returns and face lower risk by tempering potential losses when the market is down.
- Trade & Ahead is a financial consultancy firm that provides customers with personalized investment strategies. To help better analyze stocks across different market segments, and help protect against risk that could make the portfolio vulnerable to losses, Trade & Ahead has hired our firm, and provided us with data comprising stock price and some financial indicators for a few companies listed under the New York Stock Exchange.
- Based on the data provided, we performed exploratory data and K-Means Cluster and Hierarchical Cluster analysis. We identified stocks that exhibit similar characteristics and ones which exhibit minimum correlation. In this report I will be sharing my findings by grouping the stocks based on the attributes provided, and I will be sharing insights about the characteristics of each group.

Data Background, Preprocessing, and Univariate Analysis

- The dataset we are exploring has 340 rows and 15 columns. There are no missing values or duplicates.

- The first two columns refer to the ticker symbol and name of the company. The next two columns classify the stocks per sector and sub industry according to the Global Industry Classification Standard.

- The remaining 11 columns are numeric in nature. Univariate analysis on these features reveal the presence of extreme outliers, and normal to right skewed distributions for most.

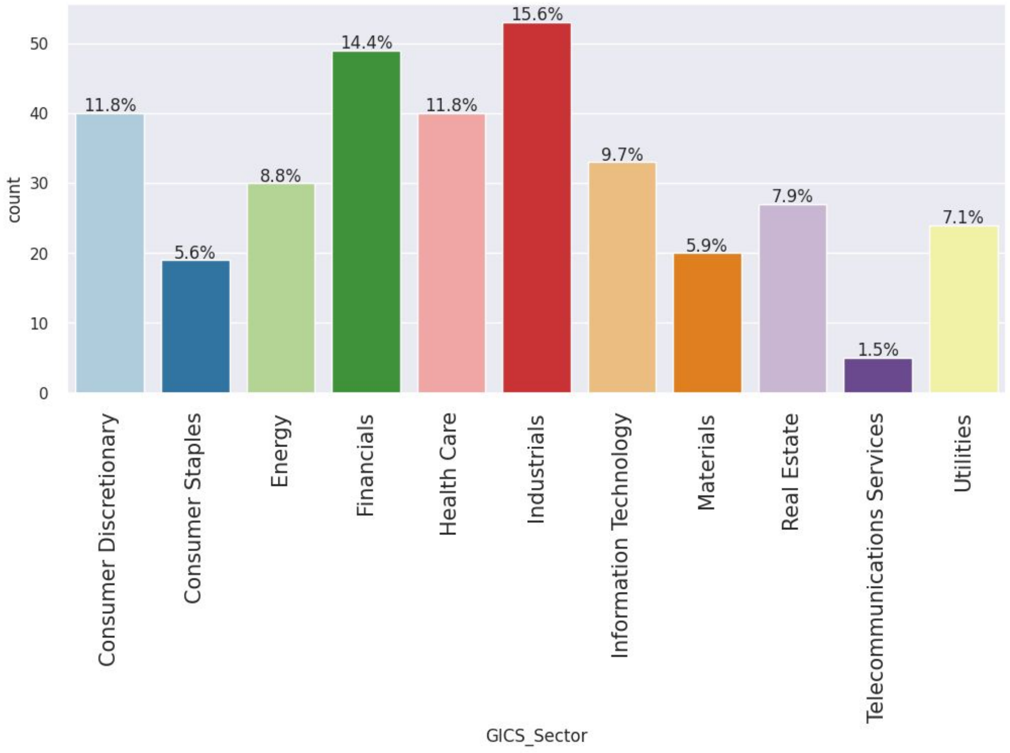
```
# checking the column names and datatypes
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 340 entries, 0 to 339
Data columns (total 15 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   Ticker Symbol                 340 non-null    object
 1   Security                      340 non-null    object
 2   GICS Sector                   340 non-null    object
 3   GICS Sub Industry             340 non-null    object
 4   Current Price                 340 non-null    float64
 5   Price Change                  340 non-null    float64
 6   Volatility                    340 non-null    float64
 7   ROE                           340 non-null    int64
 8   Cash Ratio                    340 non-null    int64
 9   Net Cash Flow                 340 non-null    int64
 10  Net Income                    340 non-null    int64
 11  Earnings Per Share            340 non-null    float64
 12  Estimated Shares Outstanding  340 non-null    float64
 13  P/E Ratio                     340 non-null    float64
 14  P/B Ratio                     340 non-null    float64
dtypes: float64(7), int64(4), object(4)
memory usage: 40.0+ KB
```

Link to Appendix slide on data background check

Bivariate Analysis

```
labeled_barplot(df, 'GICS_Sector', perc=True)
```

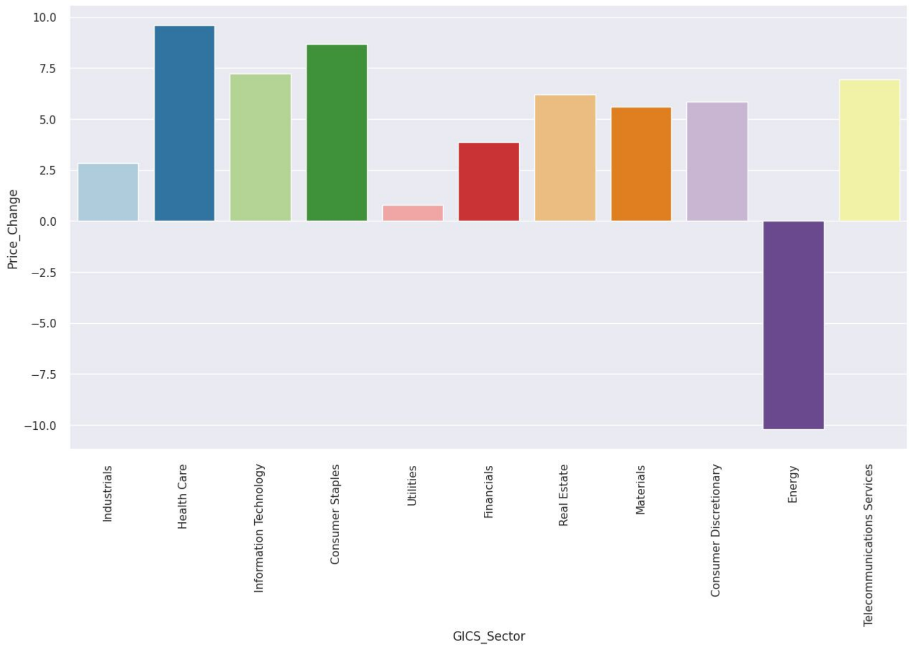


Observation:

- Industrials and Financials represent 30% of the data, followed by Consumer Discretionary and Healthcare at 11.8% each.
- Telecommunication Services only represents 1.5% of the data

Bivariate Analysis

```
plt.figure(figsize=(15,8)) sns.barplot(data=df, x='GICS_Sector',
y='Price_Change', ci=False, palette="Paired") ## Complete the code
 to choose the right variables  plt.xticks(rotation=90) plt.show()
```



Observation:

The stocks of the following economic sectors have seen the maximum price increase on average:

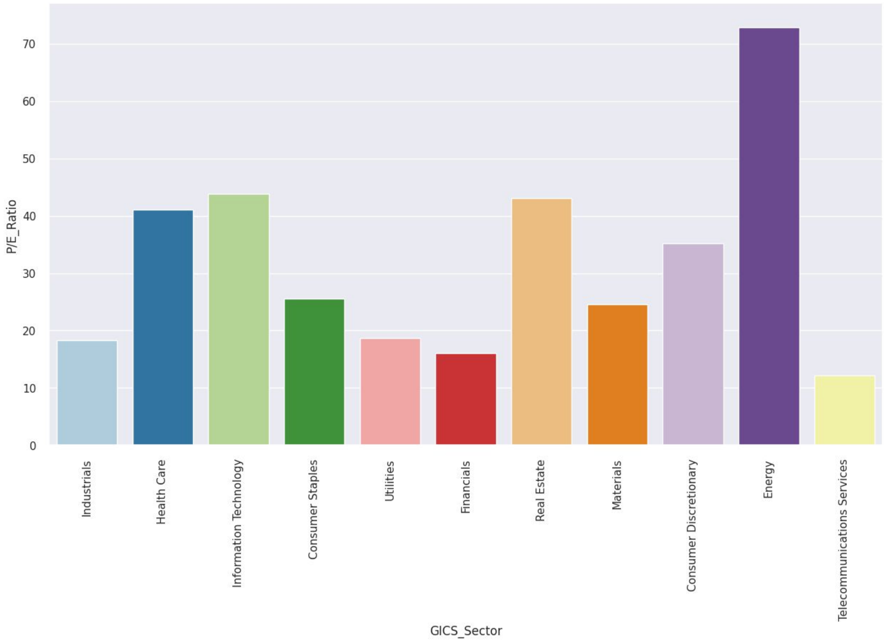
- Health Care
- Consumer Staples
- Information Technology
- Telecommunication Services

In contrast, the following sector has seen a dramatic price decrease on average:

- Energy sector

Bivariate Analysis

```
plt.figure(figsize=(15,8))sns.barplot(data=df,
x='GICS_Sector', y='P/E_Ratio', ci=False, palette='Paired')
## Complete the code to choose the right variables
plt.xticks(rotation=90) plt.show()
```

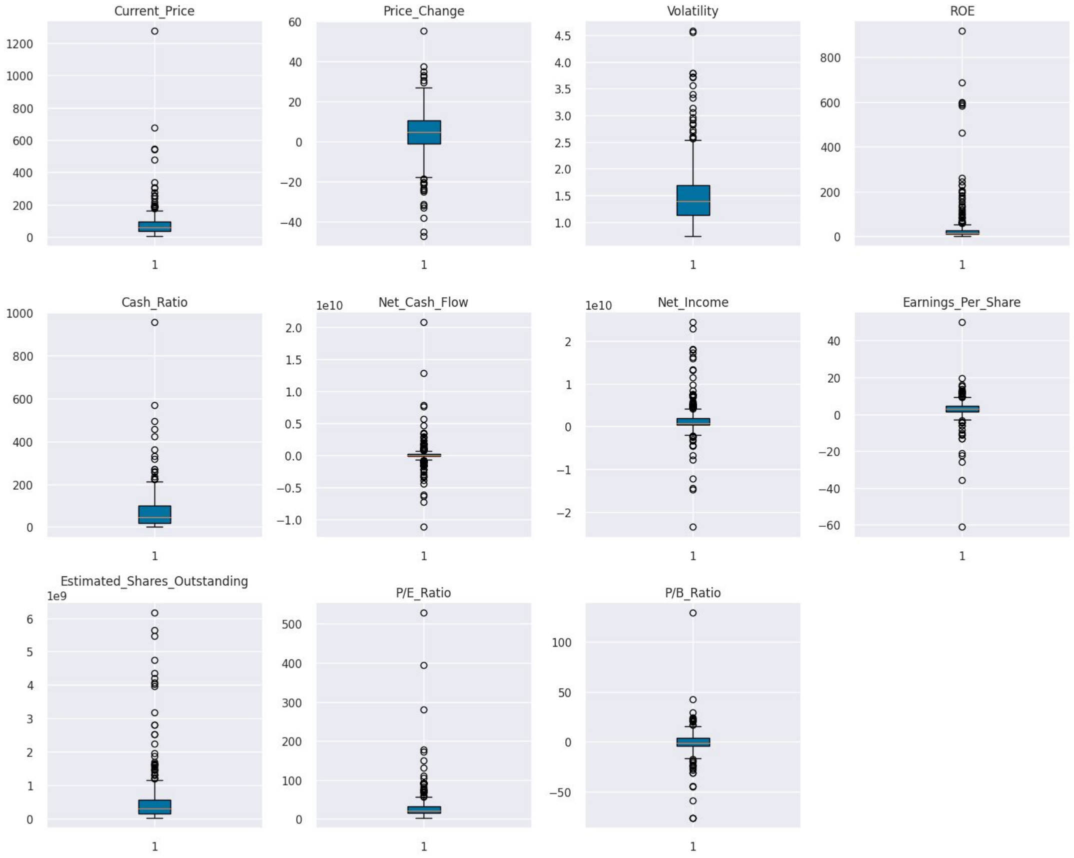


Observation:

Economy Sectors with highest Price to Earnings Ratio (the amount an investor is willing to invest in a single share of a company per dollar of its earnings):

- Energy (with head and shoulders on top)
- Information Technology
- Real Estate
- Health Care

Data Preprocessing - outlier treatment and scaling

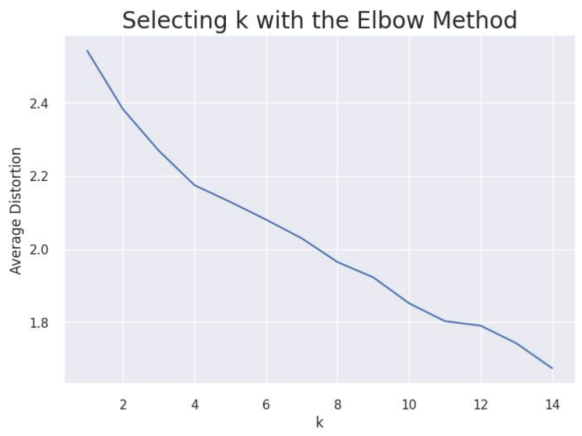


Observation:

- To address the extreme outliers, we apply a whisker limit of 1.5 to all columns.
- Most features have right skewed and normal distributions.
- Before we can proceed with the clustering analysis, we need to scale the data needs to a normal distribution to address the wide variety of values between (and within) the various columns of the dataset.

Link to Appendix slide on Outlier Treatment

K-Means Clustering Summary - optimal number of clusters




```
Number of Clusters: 1   Average Distortion: 2.5425069919221697
Number of Clusters: 2   Average Distortion: 2.382318498894466
Number of Clusters: 3   Average Distortion: 2.2692367155390745
Number of Clusters: 4   Average Distortion: 2.1745559827866363
Number of Clusters: 5   Average Distortion: 2.128799332840716
Number of Clusters: 6   Average Distortion: 2.080400099226289
Number of Clusters: 7   Average Distortion: 2.0289794220177395
Number of Clusters: 8   Average Distortion: 1.964144163389972
Number of Clusters: 9   Average Distortion: 1.9221492045198068
Number of Clusters: 10  Average Distortion: 1.8513913649973124
Number of Clusters: 11  Average Distortion: 1.8024134734578485
Number of Clusters: 12  Average Distortion: 1.7900931879652673
Number of Clusters: 13  Average Distortion: 1.7417609203336912
Number of Clusters: 14  Average Distortion: 1.673559857259703
```

Observation:

- The appropriate value of K from the elbow curve may be 4. It is tricky since the number of companies provided in the dataset is very limited, and it is risky to create too many clusters, yet we do need as many clusters as possible in order to extract hidden granular information.

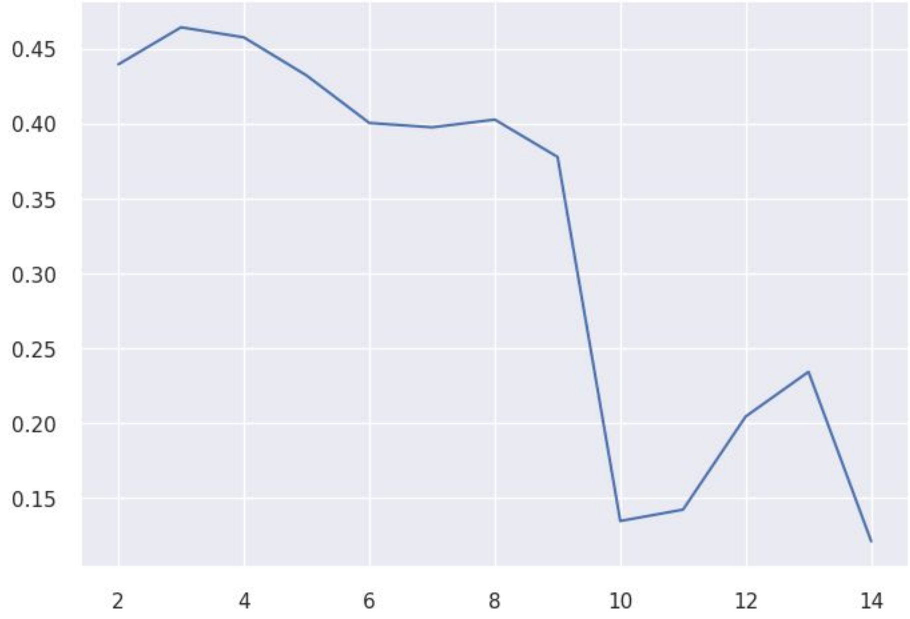
- We check the distortion score in the appendix to see if 4 clusters is a safe choice

Link to Appendix slide on K-Means Clustering

K-Means Clustering Summary - optimal number of clusters

Great Learning
POWER AHEAD

- Let’s check the silhouette scores

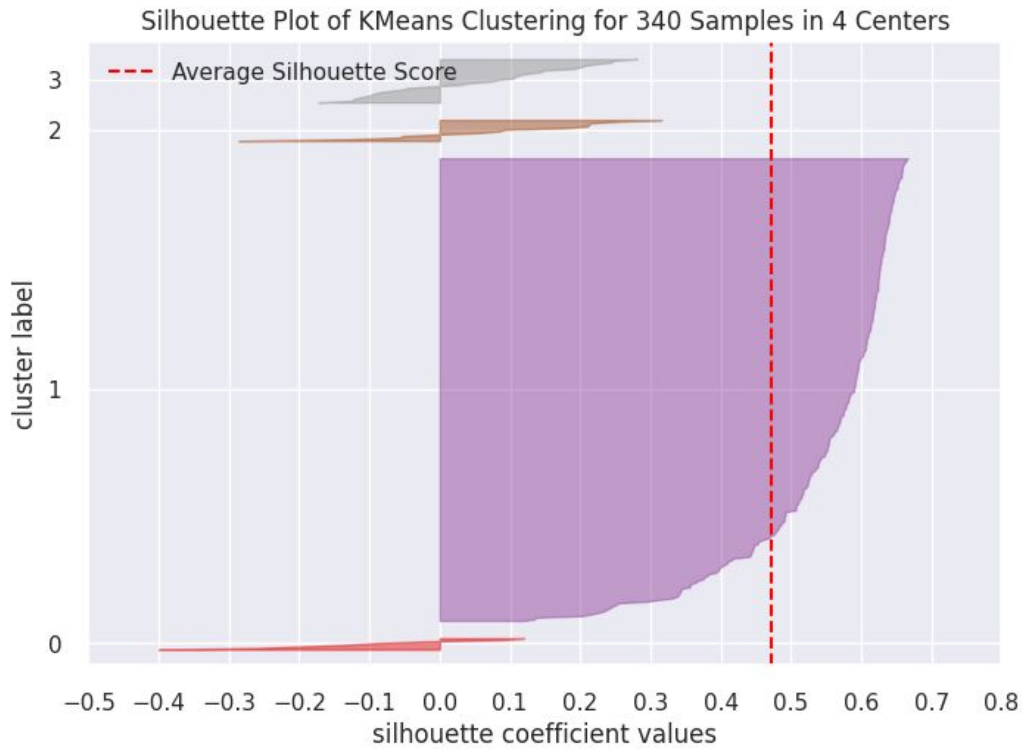



```
For n_clusters = 2, the silhouette score is 0.43969639509980457)
For n_clusters = 3, the silhouette score is 0.4644405674779404)
For n_clusters = 4, the silhouette score is 0.4577225970476733)
For n_clusters = 5, the silhouette score is 0.43228336443659804)
For n_clusters = 6, the silhouette score is 0.40054227372136l7)
For n_clusters = 7, the silhouette score is 0.3976335364987305)
For n_clusters = 8, the silhouette score is 0.40278401969450467)
For n_clusters = 9, the silhouette score is 0.3778585981433699)
For n_clusters = 10, the silhouette score is 0.13458938329968687)
For n_clusters = 11, the silhouette score is 0.1421832155528444)
For n_clusters = 12, the silhouette score is 0.2044669621527429)
For n_clusters = 13, the silhouette score is 0.23424874810104204)
For n_clusters = 14, the silhouette score is 0.12102526472829901)
```

Observation:

- It seems that 3 and 4 clusters have the highest silhouette scores. Reaffirming 4 clusters seems reasonable.

K-Means Clustering Summary - optimal number of clusters



Observation:

As with the 3 cluster plot (shown in the appendix), using 4 clusters similarly seems to create one massive cluster and three much smaller ones.

Given the choice between 3 and 4, let's go with 4, to get as granular analysis as possible.

```
# final K-means model
kmeans = KMeans(n_clusters=4, random_state=1)
## Complete the code to choose the number of clusters   kmeans.fit(k_means_df)
```

```
▼ KMeans
KMeans(n_clusters=4, random_state=1)
```

Link to Appendix slide on Outlier Treatment

K-Means Cluster Profiling

KM_segments	Current_Price	Price_Change	Volatility	ROE	Cash_Ratio	Net_Cash_Flow	Net_Income	Earnings_Per_Share	Estimated_Shares_Outstanding	P/E_Ratio	P/B_Ratio	count_in_each_segment
0	38.099260	-15.370329	2.910500	107.074074	50.037037	-159428481.481481	-3887457740.740741	-9.473704	480398572.845926	90.619220	1.342067	27
1	50.517273	5.747586	1.130399	31.090909	75.909091	-1072272727.272727	14833090909.090910	4.154545	4298826628.727273	14.803577	-4.552119	11
2	72.399112	5.066225	1.388319	34.620939	53.000000	-14046223.826715	1482212389.891697	3.621029	438533835.667184	23.843656	-3.358948	277
3	234.170932	13.400685	1.729989	25.600000	277.640000	1554926560.000000	1572611680.000000	6.045200	578316318.948800	74.960824	14.402452	25

Cluster 0

Cluster 0	Energy	22
	Industrials	1
	Information Technology	3
	Materials	1

- Stands out with high Volatility, Price_Change, ROE, and P/E_Ratio. In general this suggests high growth companies with potentially higher risk and reward. In this cluster, the majority of companies are in the Energy sector, which may have experienced extreme fluctuation in oil and gas prices.

K-Means Cluster per GICS sector

Cluster 1		
	Consumer Discretionary	1
	Consumer Staples	1
	Energy	1
	Financials	3
	Health Care	2
	Information Technology	1
	Telecommunications Services	2

- Stands out in categories Net_Income and Estimated_Shares_Outstanding. This suggests these are likely large, well-established companies with a mature business model.

Cluster 2		
	Consumer Discretionary	33
	Consumer Staples	17
	Energy	6
	Financials	45
	Health Care	29
	Industrials	52
	Information Technology	24
	Materials	19
	Real Estate	26
	Telecommunications Services	2
	Utilities	24

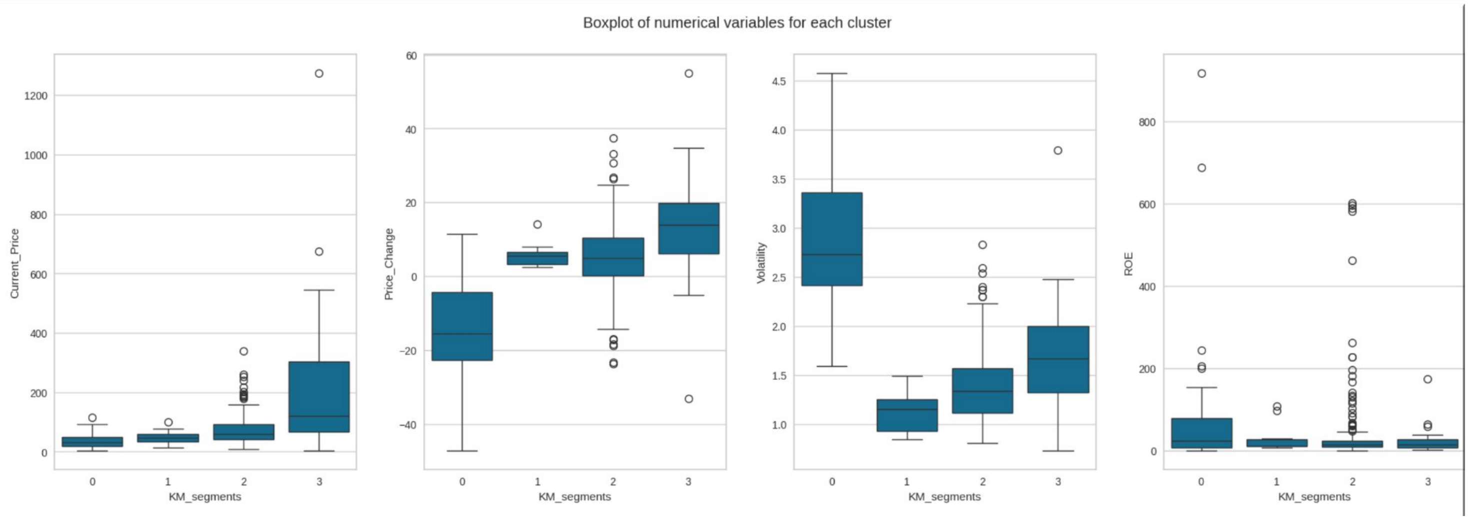
- Stands out in high number of companies in this cluster: 277

Cluster 3		
	Consumer Discretionary	6
	Consumer Staples	1
	Energy	1
	Financials	1
	Health Care	9
	Information Technology	5
	Real Estate	1
	Telecommunications Services	1

- Stands out in Current_Price, Price_Change, Cash_Ratio, Net_Cash_Flow, P/B_Ratio. To a lesser extent: Earnings_Per_Share. This paints a picture of potentially **financially strong, established companies with recent price momentum**.

Link to Appendix slide on K-means Cluster per GICS sector check

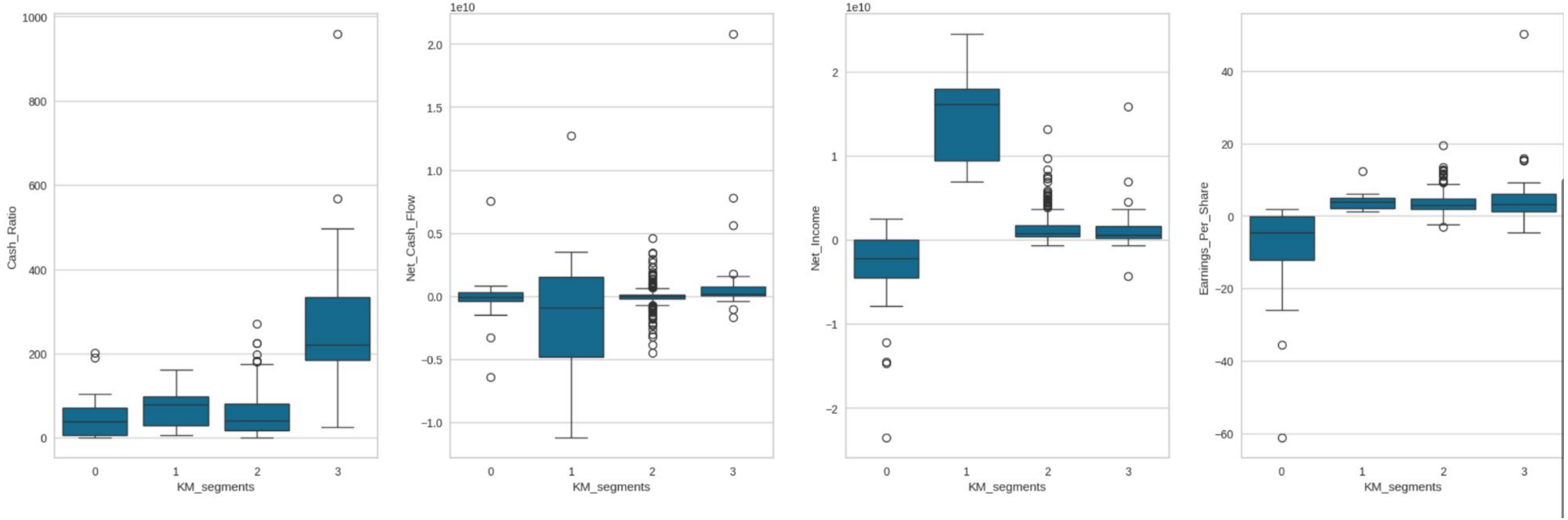
Current Price -Price Change - Volatility - ROE -for each K-Means Cluster



Observation:

- Cluster 0 stands out for its Volatility, and has its IQR in the negative digits when it comes to Price_Change.
- Cluster 1 is the least volatile and subjective to price change.

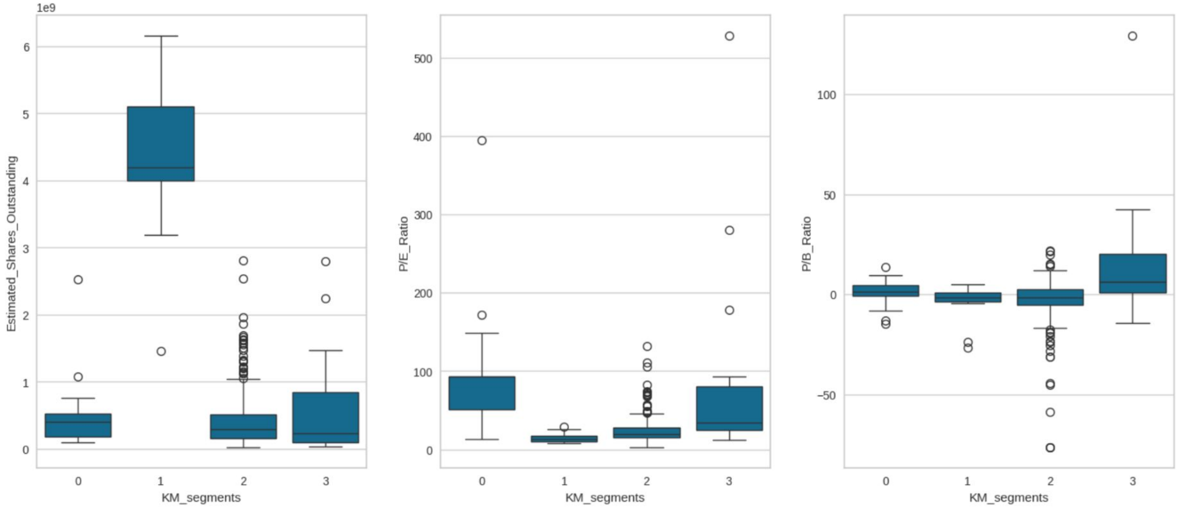
Cash Ratio - Net Cash Flow - Net Income- Earnings per Share / per K-Means Cluster



Observation:

- Cluster 1 stands out in Net_Income.
- Cluster 3 stands out in wide and high IQR in Cash_Ratio

Estimated Shares outstanding- P/E and P/B Ratios - for each K-Means Cluster



Observation:

- Cluster 1 stands out Estimated Shares Outstanding.
- No major distinctions in P/E and P/B ratios.

Hierarchical Structuring: creating agglomerative clustering model

	Linkage	Cophenetic Coefficient
4	ward	0.710118
1	complete	0.787328
5	weighted	0.869378
0	single	0.923227
3	centroid	0.931401
2	average	0.942254

Observation:

Based on clarity of dendrogram (see link appendix) and high cophenetic coefficient score, we chose average linkage method to develop a hierarchical clustering model.

Creating model using sklearn:

```
HCmodel = AgglomerativeClustering(n_clusters=6, affinity="euclidean", linkage="average")  ## Complete the code to define the hierarchical clustering model
HCmodel.fit(hc_df)
```

```
▼ AgglomerativeClustering
AgglomerativeClustering(affinity='euclidean', linkage='average', n_clusters=6)
```

Link to Appendix slide on Dendrogram

Hierarchical Clustering Profile Summary

HC_segments	Current_Price	Price_Change	Volatility	ROE	Cash_Ratio	Net_Cash_Flow	Net_Income	Earnings_Per_Share
0	77.287589	4.099730	1.518066	35.336336	66.900901	-33197321.321321	1538074666.666667	2.885270
1	25.640000	11.237908	1.322355	12.500000	130.500000	16755500000.000000	13654000000.000000	3.295000
2	24.485001	-13.351992	3.482611	802.000000	51.000000	-1292500000.000000	-19106500000.000000	-41.815000
3	104.660004	16.224320	1.320606	8.000000	958.000000	592000000.000000	3669000000.000000	1.310000
4	1274.949951	3.190527	1.268340	29.000000	184.000000	-1671386000.000000	2551360000.000000	50.090000
5	276.570007	6.189286	1.116976	30.000000	25.000000	90885000.000000	596541000.000000	8.910000

Estimated_Shares_Outstanding	P/E_Ratio	P/B_Ratio	count_in_each_segment
560505037.293544	32.441706	-2.174921	333
2791829362.100000	13.649696	1.508484	2
519573983.250000	60.748608	1.565141	2
2800763359.000000	79.893133	5.884467	1
50935516.070000	25.453183	-1.052429	1
66951851.850000	31.040405	129.064585	1

[Link to Appendix slide on Hierarchical Clustering]

Hierarchical Clustering Profile Observations

Cluster 0: This cluster contains the largest number of companies with 333, out of 340 total. There are no other major signifiers for this group, so this may not be the best use of clustering that many companies together.

Cluster 1: Contains two companies, and stands out in Net_Cashflow and Net_Income.

Cluster 2: Contains two companies, and stands out in Volatility and ROE.

Cluster 3: Comprises of one company that stands out in Price_Change Cash_Ratio, and Price_Change.

Cluster 4: Comprises of one company that stands out in Current_Price Earnings_Per_Share, Estimated_Shares_Outstanding, and P/E Ratio.

Cluster 5: Comprises of one company and stands out for P/B Ratio.

```
In cluster 5, the following companies are present:
['Alliance Data Systems']

In cluster 2, the following companies are present:
['Apache Corporation' 'Chesapeake Energy']

In cluster 1, the following companies are present:
['Bank of America Corp' 'Intel Corp.']

In cluster 3, the following companies are present:
['Facebook']

In cluster 4, the following companies are present:
['Priceline.com Inc']
```

APPENDIX

Data Background / Preprocessing - missing values /duplicates

```
## Complete the code to import the data
data =
pd.read_csv('/content/drive/MyDrive/Unsupervised
Learning/stock_data.csv')
```

```
# checking shape of the data
shape = data.shape
print(f"There are {shape[0]} rows and {shape[1]}
columns.")
## Complete the code to get the shape of data
```

```
There are 340 rows and 15 columns.
```

```
# checking for duplicate values
df.duplicated().sum()
## Complete the code to get total number of duplicate
values
```

```
0
```

```
# checking for missing values in the data
df.isnull().sum()
## Complete the code to check the missing values in
the data
```

```
Ticker Symbol                  0
Security                       0
GICS Sector                    0
GICS Sub Industry              0
Current Price                  0
Price Change                   0
Volatility                     0
ROE                            0
Cash Ratio                     0
Net Cash Flow                  0
Net Income                     0
Earnings Per Share             0
Estimated Shares Outstanding   0
P/E Ratio                      0
P/B Ratio                      0
dtype: int64
```

Data Background and Contents

```
#let's clean up the column names
df.columns = [c.replace(" ", "_") for c in df.columns]
print('New column names:\n', df.columns)
```

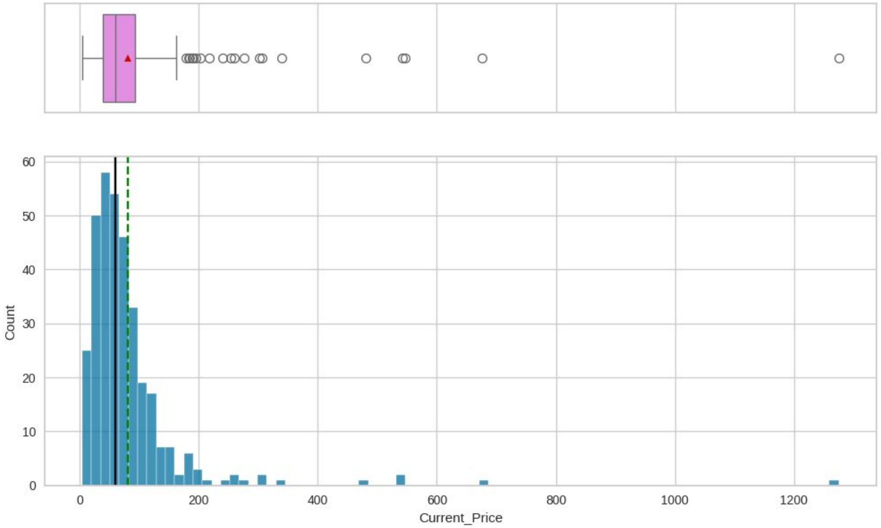
```
df.describe(include='all').T
```

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
Ticker_Symbol	340	340	AAL	1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Security	340	340	American Airlines Group	1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
GICS_Sector	340	11	Industrials	53	NaN	NaN	NaN	NaN	NaN	NaN	NaN
GICS_Sub_Industry	340	104	Oil & Gas Exploration & Production	16	NaN	NaN	NaN	NaN	NaN	NaN	NaN
Current_Price	340.0	NaN	NaN	NaN	80.862345	98.055086	4.5	38.555	59.705	92.880001	1274.949951
Price_Change	340.0	NaN	NaN	NaN	4.078194	12.006338	-47.129693	-0.939484	4.819505	10.695493	55.051683
Volatility	340.0	NaN	NaN	NaN	1.525976	0.591798	0.733163	1.134878	1.385593	1.695549	4.580042
ROE	340.0	NaN	NaN	NaN	39.597059	96.547538	1.0	9.75	15.0	27.0	917.0
Cash_Ratio	340.0	NaN	NaN	NaN	70.023529	90.421331	0.0	18.0	47.0	99.0	958.0
Net_Cash_Flow	340.0	NaN	NaN	NaN	55537620.588235	1946365312.175789	-11208000000.0	-193906500.0	2098000.0	169810750.0	20764000000.0
Net_Income	340.0	NaN	NaN	NaN	1494384602.941176	3940150279.327936	-23528000000.0	352301250.0	707336000.0	1899000000.0	24442000000.0
Earnings_Per_Share	340.0	NaN	NaN	NaN	2.776662	6.587779	-61.2	1.5575	2.895	4.62	50.09
stimated_Shares_Outstanding	340.0	NaN	NaN	NaN	577028337.75403	845849595.417695	27672156.86	158848216.1	309675137.8	573117457.325	6159292035.0
P/E_Ratio	340.0	NaN	NaN	NaN	32.612563	44.348731	2.935451	15.044653	20.819876	31.764755	528.039074
P/B_Ratio	340.0	NaN	NaN	NaN	-1.718249	13.966912	-76.119077	-4.352056	-1.06717	3.917066	129.064585

Univariate Analysis

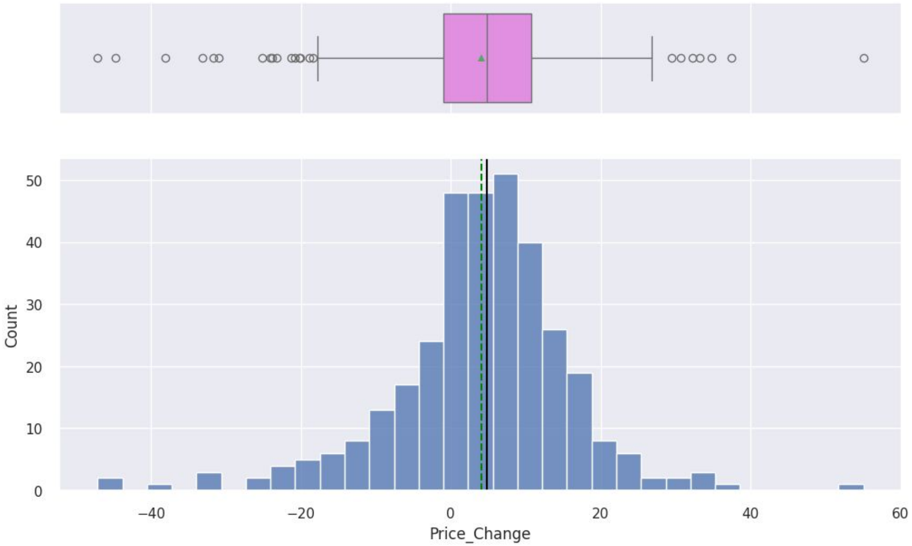
histogram_boxplot(df, 'Current_Price')

Observation: right skewed distribution
Mean of approx. $81, and median of $60
One extreme outlier at $1,300.




```
histogram_boxplot(df,'Price_Change')   ## Complete the code to create histogram_boxplot for 'Price Change'
```

Fairly normal distributed data with a mean of approx. $4.00. some outliers on either side.

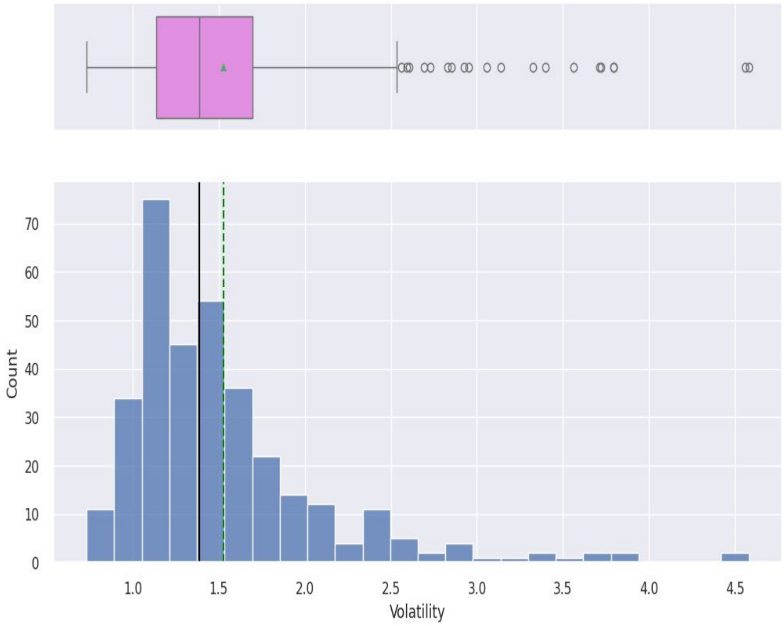


Univariate Analysis

```
histogram_boxplot(df, 'Volatility')
## Complete the code to create histogram_boxplot for 'Volatility'
```

Right skewed distribution.

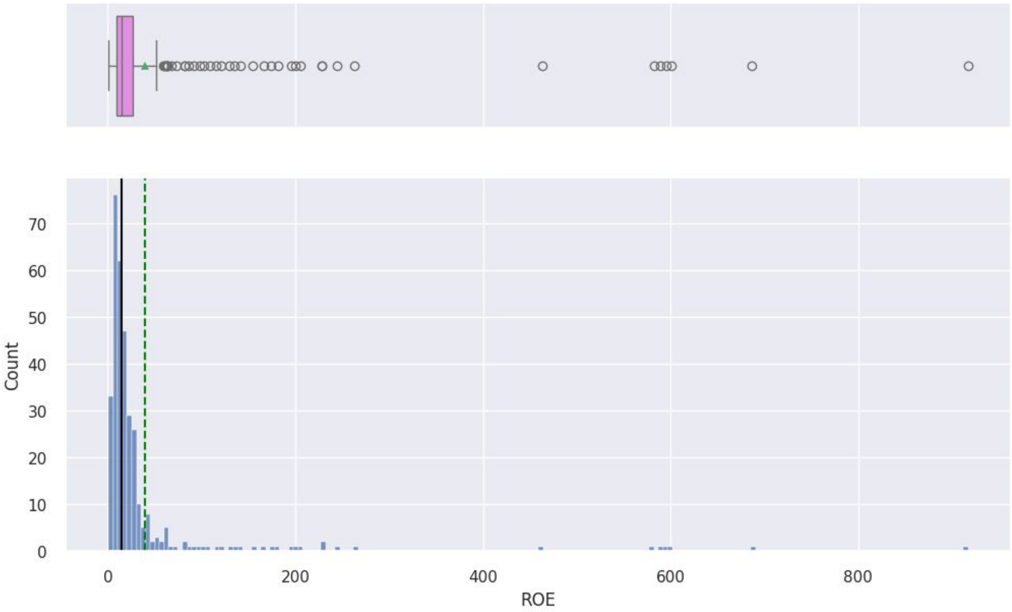
Median $1.38, and Mean of $1.38.




```
histogram_boxplot(df, 'ROE')  ## Complete the code to create histogram_boxplot for 'ROE'
```

Extreme right skewed distribution and outliers

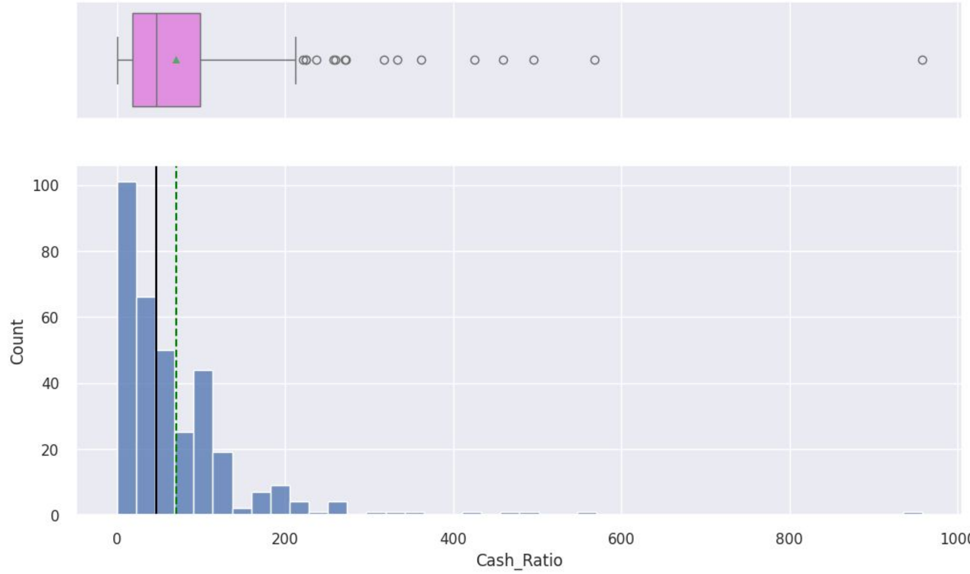
Median “return on equity” $15.00 vs. Mean of $40.00.



Univariate Analysis

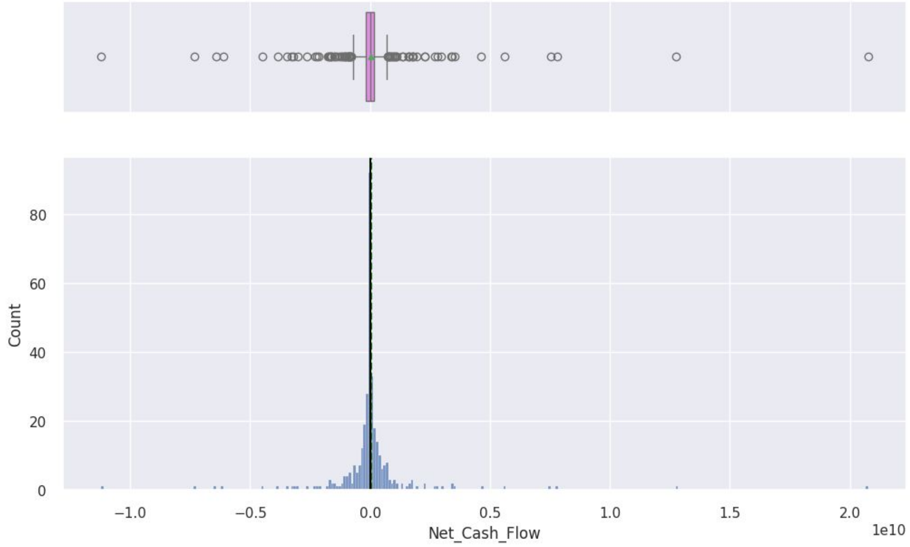
```
histogram_boxplot(df,'Cash_Ratio')   ## Complete the code to create histogram_boxplot for 'Cash Ratio'
```

Right skewed distribution. Mean $70, Median $47.




```
histogram_boxplot(df, 'Net_Cash_Flow')   ## Complete the code to create histogram_boxplot for 'Net Cash Flow'
```

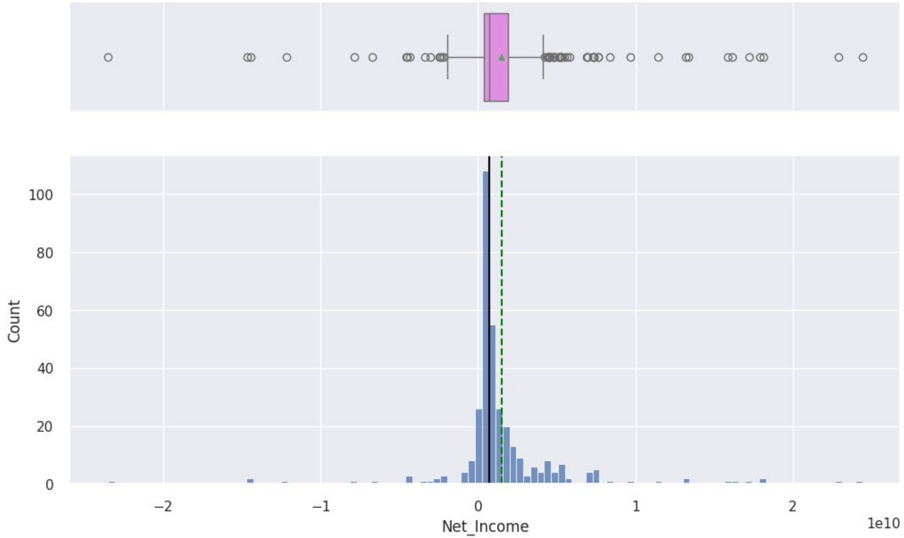
Normally distributed data but with extreme long tails on both sides. Mean 55537620.588235 Median 2098000.0



Univariate Analysis

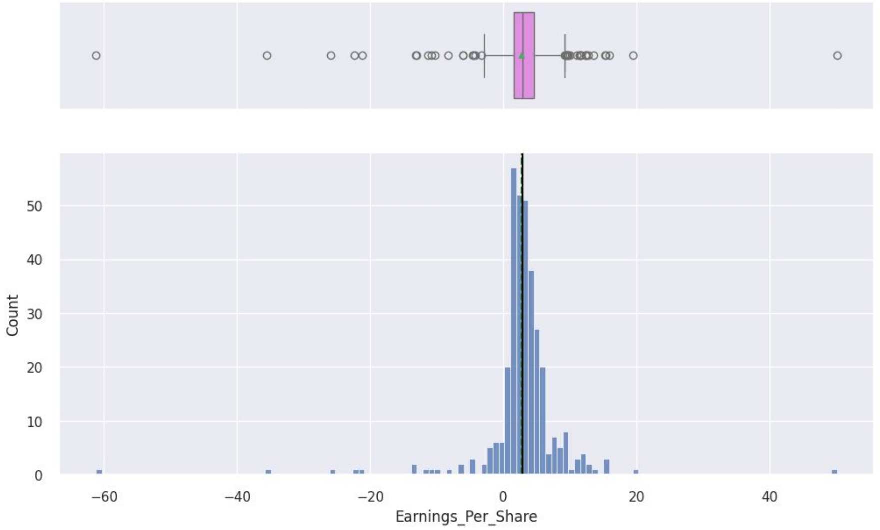
```
histogram_boxplot(df,'Net_Income')
## Complete the code to create histogram_boxplot
for 'Net_Income'
```

Extreme long tails on slightly right skewed distribution. Mean 1494384602.941176, Median 707336000.0




```
histogram_boxplot(df, 'Earnings_Per_Share')  ##
Complete the code to create histogram_boxplot for
'Earnings_Per_Share'
```

Slightly left skewed, otherwise normal distribution with long tails on both sides. Mean $2.7 / Median $2.8

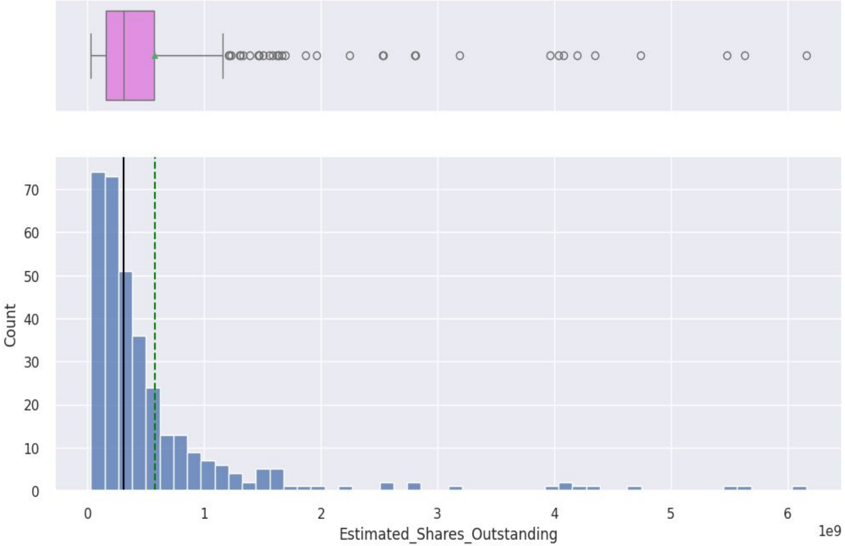


Univariate Analysis

```
histogram_boxplot(df, 'Estimated_Shares_Outstanding')  ## Complete the code to create histogram_boxplot for 'Estimated Shares Outstanding'
```

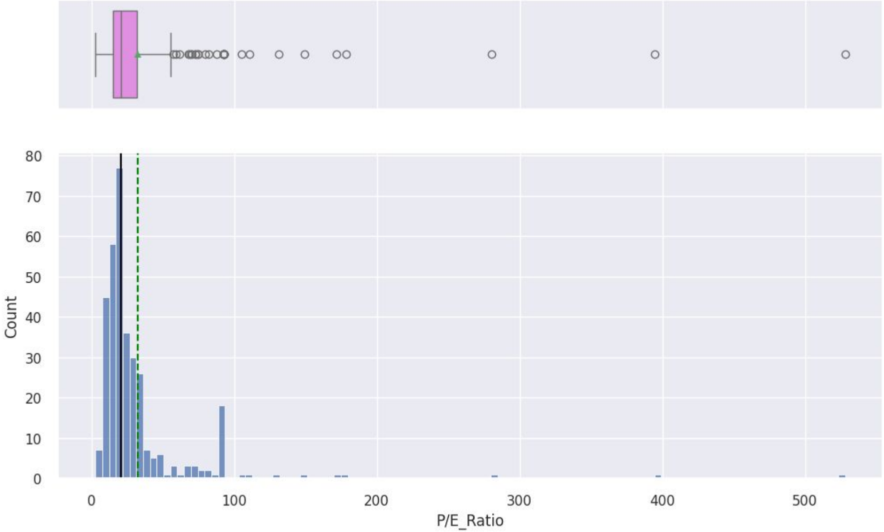
Extreme right skewed distribution.

Mean 577028337.75403 Median 309675137.8




```
histogram_boxplot(df, 'P/E_Ratio')  ## Complete the code to create histogram_boxplot for 'P/E Ratio'
```

Right skewed distribution with mean $33 and Median $21.

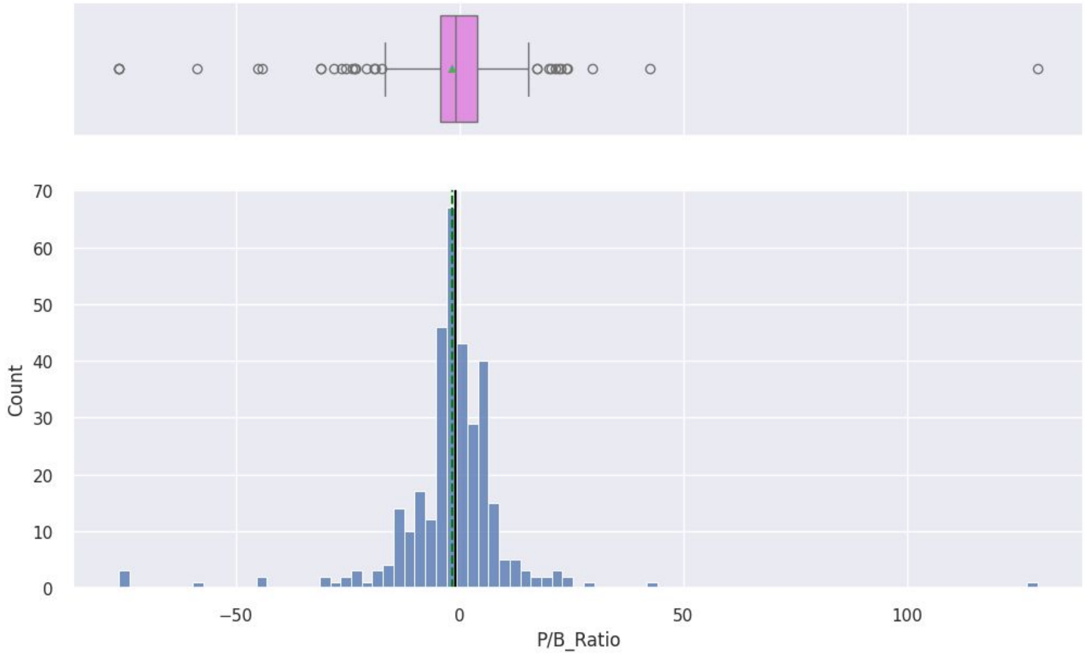


Univariate Analysis

```
histogram_boxplot(df, 'P/B_Ratio')  ## Complete the code to create histogram_boxplot for 'P/B_Ratio'
```

Fairly normal distributed with long tails on either side.

Mean is -1.718249 Median is -1.06717

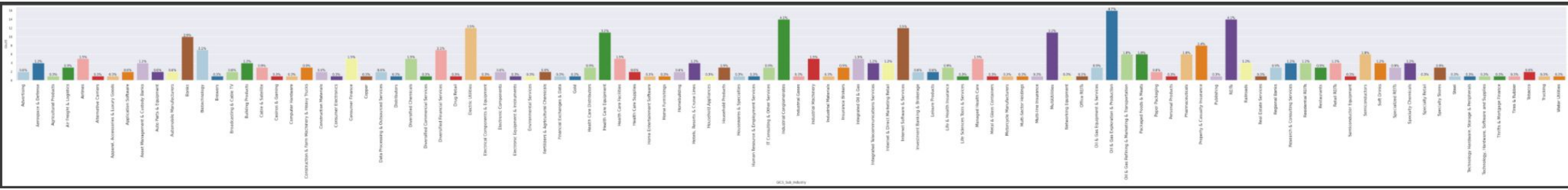


Bivariate Analysis

```
labeled_barplot(df, 'GICS_Sub_Industry', perc=True)   ## Complete the code to create a labelled barplot 
for 'GICS_Sub_Industry'
```

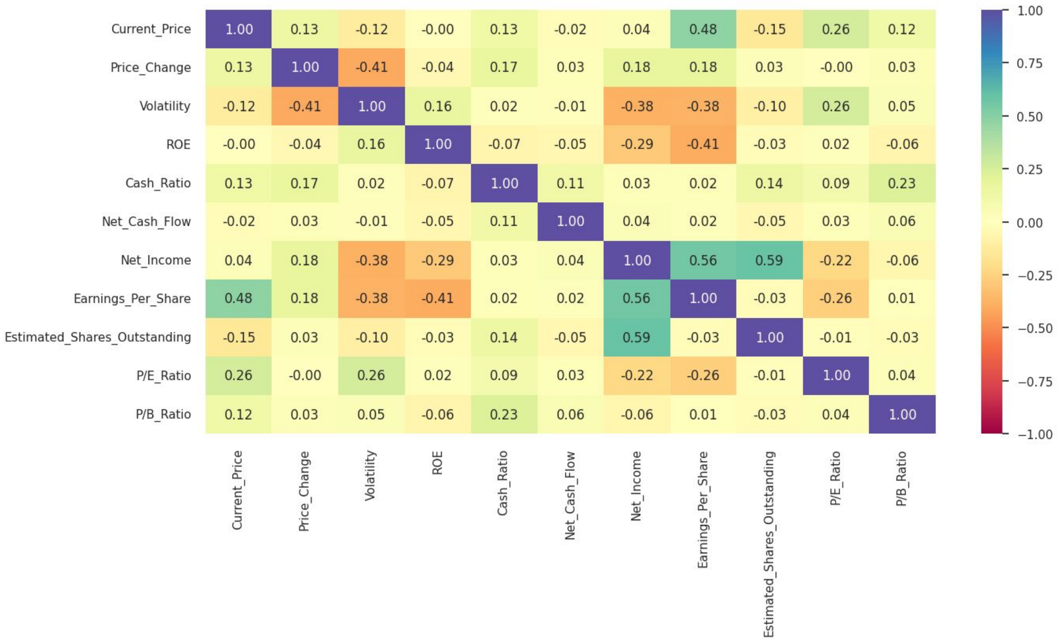
Observation: Largest sub industries represented are:

- Oil & Gas Exploration & Production (4.7%)
- Industrial Conglomerates (4.1%)
- REITs (Real Estate Investment Trust) (4.1%)
- Internet Software & Services (3.5%)
- Electric Utilities (3.5%)
- Health Care Equipment (3.2%)
- Multi Utilities (3.2%)



Bivariate Analysis

```
# correlation check plt.figure(figsize=(15, 7))
sns.heatmap(    df[num_col].corr(), annot=True,
vmin=-1, vmax=1, fmt=".2f", cmap="Spectral")
plt.show()
```

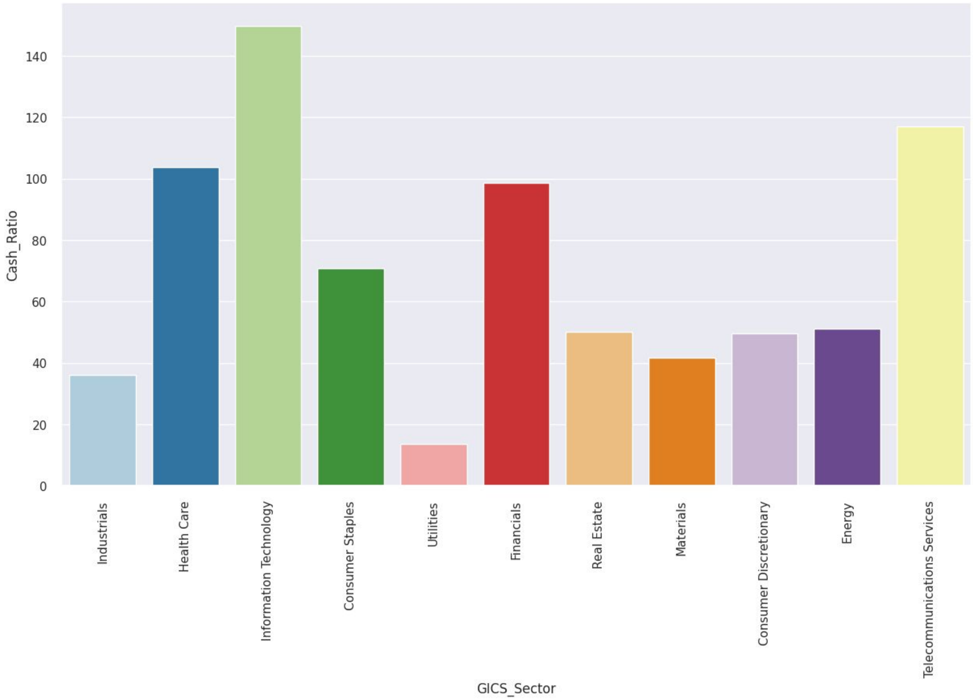


Observation:

- Net_Income is positively correlated with Earnings_Per_Share, and Estimated_Shares_Outstanding.
- Current_Price and Earnings_Per_Share are positively correlated.
- Volatility is negatively correlated with Earnings_Per_Share, Net_Income and Price_Change

Bivariate Analysis

```
plt.figure(figsize=(15,8)) sns.barplot(data=df, x='GICS_Sector',
y='Cash_Ratio', ci=False, palette='Paired')  ## Complete the code
to choose the right variablesplt.xticks(rotation=90) plt.show()
```



Observation:

The following economic sectors have the highest cash ratios (the ability to cover short-term obligations using only cash / cash equivalents:

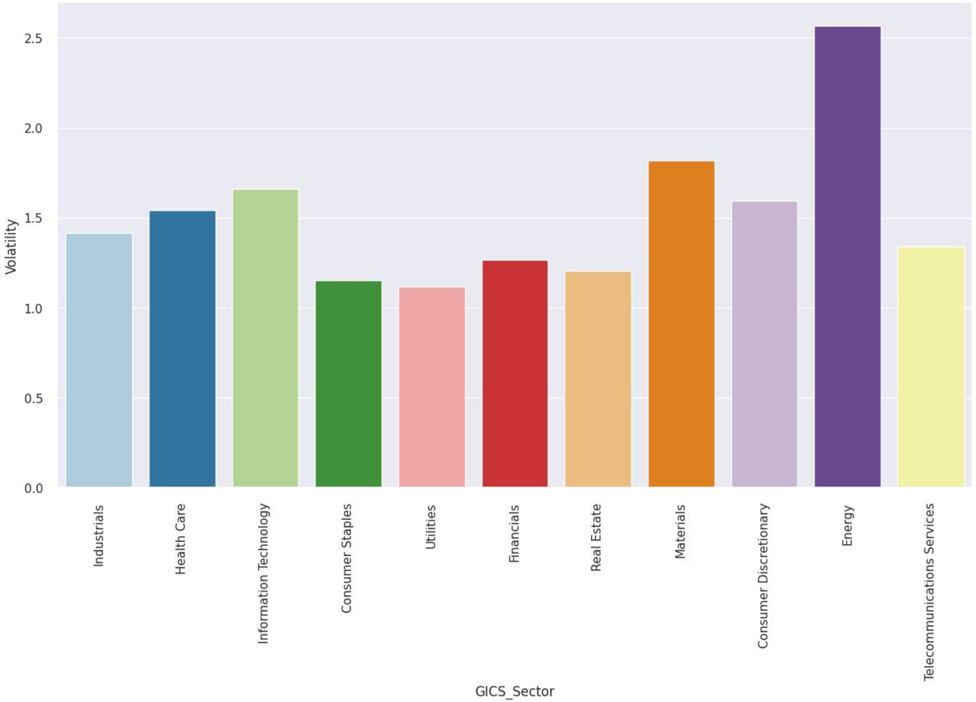
- Information Technology
- Telecommunications Services
- Health Care
- Financials

The economy sector with lowest cash ratio:

- Utilities

Bivariate Analysis

```
plt.figure(figsize=(15,8))sns.barplot(data=df, x='GICS_Sector',
y='Volatility', ci=False, palette='Paired')  ## Complete the code
to choose the right variablesplt.xticks(rotation=90)plt.show()
```



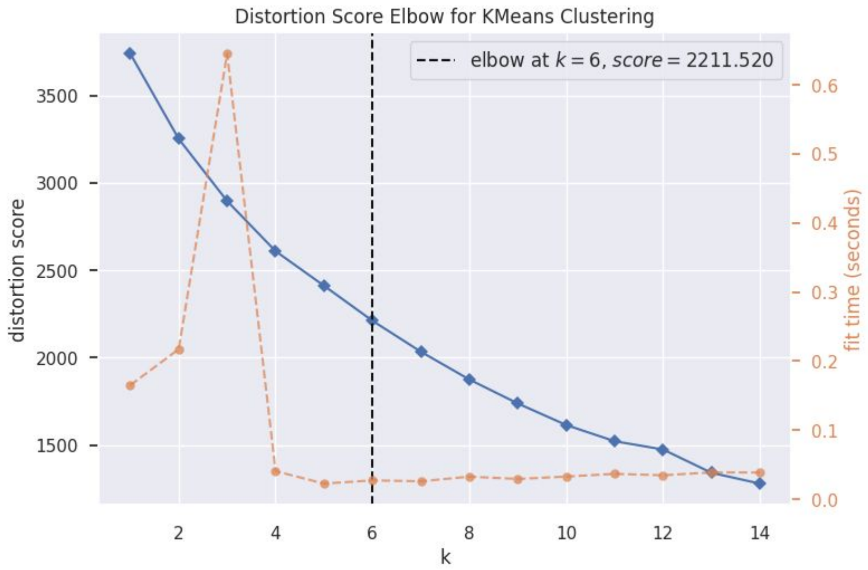
Observation:

As previous graphs already indicate, The Energy sector is by far the most volatile of all sectors.

Data Preprocessing - outlier treatment and scaling

```
plt.figure(figsize=(15, 12))
for i, variable in enumerate(num_col):
    plt.subplot(3, 4, i + 1)
    plt.boxplot(df[variable], whis=1.5, patch_artist=True)
    plt.tight_layout()
    plt.title(variable)
plt.show()

# scaling the data before clustering
scaler = StandardScaler()
subset = df[num_col].copy()  ## Complete the code to scale the data
subset_scaled = scaler.fit_transform(subset)

# creating a dataframe of the scaled data
subset_scaled_df = pd.DataFrame(subset_scaled, columns=subset.columns)
```

K-Means Clustering Technique - distortion score elbow method



Observation:

Distortion score calculates that starting with 6 clusters the information stored within the segments becomes ‘distorted’. Any cluster below 6 should still extract ‘undistorted’ information. It seems 4 clusters is a safe option.

```
Number of Clusters: 1   Average Distortion: 2.5425069919221697
Number of Clusters: 2   Average Distortion: 2.382318498894466
Number of Clusters: 3   Average Distortion: 2.2692367155390745
Number of Clusters: 4   Average Distortion: 2.1745559827866363
Number of Clusters: 5   Average Distortion: 2.128799332840716
Number of Clusters: 6   Average Distortion: 2.080400099226289
Number of Clusters: 7   Average Distortion: 2.0289794220177395
Number of Clusters: 8   Average Distortion: 1.964144163389972
Number of Clusters: 9   Average Distortion: 1.9221492045198068
Number of Clusters: 10  Average Distortion: 1.8513913649973124
Number of Clusters: 11  Average Distortion: 1.8024134734578485
Number of Clusters: 12  Average Distortion: 1.7900931879652673
Number of Clusters: 13  Average Distortion: 1.7417609203336912
Number of Clusters: 14  Average Distortion: 1.673559857259703
```

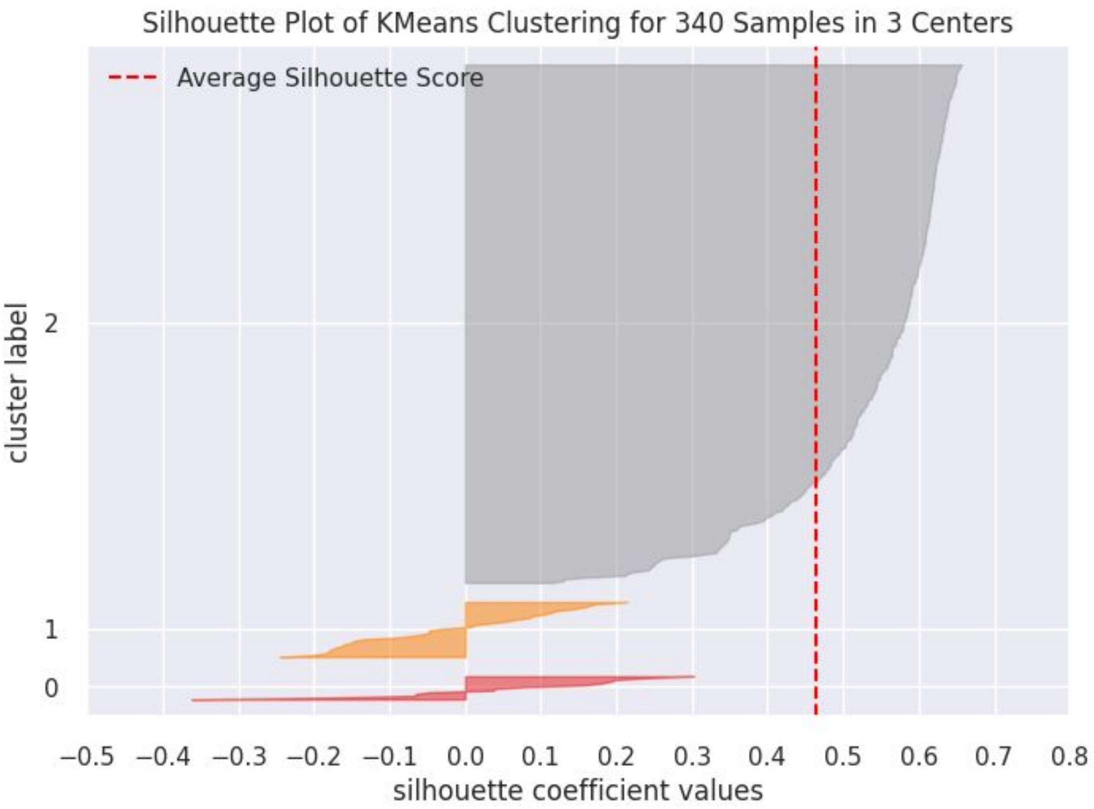
K-Means Clustering Summary - silhouette visualization

Observation:

It seems that the range of 2 to 5 number of clusters have a high enough silhouette score, which would imply that this range of clusters seems to be the most promising.

```
model = KMeans(random_state=1)
visualizer = KElbowVisualizer(model, k=(1, 15), timings=True)
visualizer.fit(k_means_df)  # fit the data to the visualizer
visualizer.show()  # finalize and render figure
```

K-Means Clustering Summary - optimal number of clusters



Observation:

Using three clusters seems to create one massive cluster and two smaller ones.

```
# finding optimal no. of clusters with silhouette coefficients
visualizer = SilhouetteVisualizer(KMeans(3, random_state=1))  ## Complete the code to visualize the silhouette scores for certain number of clusters
visualizer.fit(k_means_df)
visualizer.show()
```

K-Means Clustering Technique - silhouette score

```
sil_score = []
cluster_list = range(2, 15)
for n_clusters in cluster_list:
    clusterer = KMeans(n_clusters=n_clusters, random_state=1)
    preds = clusterer.fit_predict((subset_scaled_df))
    score = silhouette_score(k_means_df, preds)
    sil_score.append(score)
    print("For n_clusters = {}, the silhouette score is {})".format(n_clusters, score))
plt.plot(cluster_list, sil_score)
plt.show()
```

```
visualizer = SilhouetteVisualizer(KMeans(4, random_state=1))  ## Complete the code to visualize the silhouette scores for certain number of clusters
visualizer.fit(k_means_df)
visualizer.show()
```

K-Means Cluster Profiling

```
#Figure out how to have the mean apply only to numeric columns.
km_cluster_profile = df1.groupby("KM_segments")[num_col].mean()
## Complete the code to groupby the cluster labels

km_cluster_profile["count_in_each_segment"] =
(df1.groupby("KM_segments")["Security"].count().values
## Complete the code to groupby the cluster labels)
km_cluster_profile.style.highlight_max(color="lightgreen", axis=0)

## Complete the code to print the companies in each cluster
for cl in df1["KM_segments"].unique():
    print("In cluster {}, the following companies are present:".format(cl))
    print(df1[df1["KM_segments"] == cl]["Security"].unique())
    print()

df1.groupby(["KM_segments", "GICS_Sector"])['Security'].count()
```


```
## Complete the code to print the
companies in each cluster
for cl in df1["KM_segments"].unique():
    print("In cluster {}, the following
companies are present:".format(cl))
    print(df1[df1["KM_segments"] ==
cl]["Security"].unique())
    print()
```

K-Means Cluster 2: individual companies

```
In cluster 2, the following companies are present
['American Airlines Group' 'AbbVie' 'Abbott Laboratories''Adobe Systems Inc' 'Archer-Daniels-Midland Co'
'Ameren Corp' 'American Electric Power' 'AFLAC Inc''American International Group, Inc.' 'Apartment
Investment & Mgmt' 'Assurant Inc' 'Arthur J. Gallagher & Co.' 'Akamai Technologies Inc''Albemarle Corp'
'Alaska Air Group Inc' 'Allstate Corp' 'Allegion' 'Applied Materials Inc' 'AMETEK Inc' 'Affiliated
Managers Group Inc' 'Ameriprise Financial' 'American Tower Corp A' 'AutoNation Inc' 'Anthem Inc.' 'Aon
plc' 'Amphenol Corp' 'Arconic Inc' 'Activision Blizzard' 'AvalonBay Communities, Inc.' 'Broadcom'
 'American Water Works Company Inc' 'American Express Co' 'Boeing Company' 'Baxter International Inc.'
'BB&T Corporation' 'Bard (C.R.) Inc.''The Bank of New York Mellon Corp.' 'Ball Corp' 'Bristol-Myers
Squibb' 'Boston Scientific' 'BorgWarner' 'Boston Properties' 'Caterpillar Inc.' 'Chubb Limited' 'CBRE
Group' 'Crown Castle International Corp.''Carnival Corp.' 'CF Industries Holdings Inc' 'Citizens
Financial Group''Church & Dwight' 'C. H. Robinson Worldwide' 'Charter Communications' 'CIGNA Corp.'
'Cincinnati Financial' 'Colgate-Palmolive' 'Comerica Inc.' 'CME Group Inc.' 'Cummins Inc.' 'CMS Energy'
'Centene Corporation' 'CenterPoint Energy' 'Capital One Financial' 'The Cooper Companies' 'CSX Corp.'
'CenturyLink Inc' 'Cognizant Technology Solutions' 'Citrix Systems' 'CVS Health' 'Chevron Corp.'
'Dominion Resources' 'Delta Air Lines' 'Du Pont (E.I.)' 'Deere & Co.''Discover Financial Services' 'Quest
Diagnostics' 'Danaher Corp.' 'The Walt Disney Company' 'Discovery Communications-A' 'Discovery
Communications-C' 'Delphi Automotive' 'Digital Realty Trust''Dun & Bradstreet' 'Dover Corp.' 'Dr Pepper
Snapple Group' 'Duke Energy''DaVita Inc.' 'eBay Inc.' 'Ecolab Inc.' 'Consolidated Edison''Equifax Inc.'
"Edison Int'l" 'Eastman Chemical' 'Equity Residential''Eversource Energy' 'Essex Property Trust, Inc.' '
```

K-Means Cluster 2: individual companies continued

```
'E*Trade' 'Eaton Corporation' 'Entergy Corp.' 'Exelon Corp.' "Expeditors Int'l" 'Expedia Inc.' 'Extra
Space Storage' 'Fastenal Co' 'Fortune Brands Home & Security' 'FirstEnergy Corp' 'Fidelity National
Information Services' 'Fiserv Inc' 'FLIR Systems''Fluor Corp.' 'Flowserve Corporation' 'FMC
Corporation''Federal Realty Investment Trust' 'General Dynamics' 'General Growth Properties Inc.'
'Corning Inc.' 'General Motors' 'Genuine Parts' 'Garmin Ltd.' 'Goodyear Tire & Rubber' 'Grainger (W.W.)
Inc.' 'Hasbro Inc.' 'Huntington Bancshares' 'HCA Holdings' 'Welltower Inc.' 'HCP Inc.' 'Hartford
Financial Svc.Gp.' 'Harley-Davidson' "Honeywell Int'l Inc." 'HP Inc.' 'Hormel Foods Corp.' 'Henry Schein'
'Host Hotels & Resorts' 'The Hershey Company''Humana Inc.' 'International Business Machines' 'IDEXX
Laboratories' 'Intl Flavors & Fragrances' 'International Paper' 'Interpublic Group''Iron Mountain
Incorporated' 'Illinois Tool Works' 'Invesco Ltd.''J. B. Hunt Transport Services' 'Jacobs Engineering
Group' 'Juniper Networks' 'Kimco Realty' 'Kimberly-Clark' 'Kansas City Southern''Leggett & Platt' 'Lennar
Corp.' 'Laboratory Corp. of America Holding' 'LKQ Corporation' 'L-3 Communications Holdings' 'Lilly (Eli)
& Co.' 'Lockheed Martin Corp.' 'Alliant Energy Corp' 'Leucadia National Corp.' 'Southwest Airlines'
'Level 3 Communications' 'LyondellBasell''Mastercard Inc.' 'Mid-America Apartments' 'Macerich' "Marriott
Int'l." 'Masco Corp.' 'Mattel Inc.' "Moody's Corp" 'Mondelez International' 'MetLife Inc.' 'Mohawk
Industries' 'Mead Johnson' 'McCormick & Co.''Martin Marietta Materials' 'Marsh & McLennan' '3M Company
'Altria Group Inc' 'The Mosaic Company' 'Marathon Petroleum''Merck & Co.' 'M&T Bank Corp.' 'Mettler
Toledo' 'Mylan N.V.' 'Navient' 'NASDAQ OMX Group' 'NextEra Energy' 'Newmont Mining Corp. (Hldg. Co.)'
 'Nielsen Holdings' 'Norfolk Southern Corp.' 'Northern Trust Corp.' 'Nucor Corp.' 'Newell Brands' 'Realty
```

K-Means Cluster 2 - final

```
Income Corporation' 'Omnicom Group' "O'Reilly Automotive" "People's United Financial" 'Pitney-Bowes'
 'PACCAR Inc.' 'PG&E Corp.' 'Public Serv. Enterprise Inc.' 'PepsiCo Inc.''Principal Financial
Group''Procter & Gamble' 'Progressive Corp.''Pulte Homes Inc.' 'Philip Morris International' 'PNC
Financial Services''Pentair Ltd.' 'Pinnacle West Capital' 'PPG Industries' 'PPL Corp.''Prudential
Financial' 'Phillips 66' 'Praxair Inc.' 'PayPal''Ryder System' 'Royal Caribbean Cruises Ltd' 'Robert Half
International''Roper Industries' 'Republic Services Inc' 'SCANA Corp' 'Charles Schwab Corporation'
'Spectra Energy Corp.' 'Sealed Air' 'Sherwin-Williams' 'SL Green Realty' 'Scripps Networks Interactive
Inc.' 'Southern Co.' 'Simon Property Group Inc' 'S&P Global, Inc.' 'Stericycle Inc' 'Sempra Energy'
'SunTrust Banks' 'State Street Corp.' 'Skyworks Solutions' 'Synchrony Financial' 'Stryker Corp.'
 'Molson Coors Brewing Company' 'Tegna, Inc.' 'Torchmark Corp.' 'Thermo Fisher Scientific' 'The Travelers
Companies Inc.' 'Tractor Supply Company' 'Tyson Foods' 'Tesoro Petroleum Co.' 'Total System Services'
'Texas Instruments' 'Under Armour' 'United Continental Holdings' 'UDR Inc' 'Universal Health Services,
Inc.' 'United Health Group Inc.' 'Unum Group' 'Union Pacific' 'United Parcel Service' 'United
Technologies' 'Varian Medical Systems' 'Valero Energy' 'Vulcan Materials' 'Vornado Realty Trust'
 'Verisk Analytics' 'Verisign Inc.' 'Ventas Inc' 'Wec Energy Group Inc' 'Whirlpool Corp.' 'Waste
Management Inc.' 'Western Union Co' 'Weyerhaeuser Corp.' 'Wyndham Worldwide' 'Xcel Energy Inc' 'XL
Capital' 'Dentsply Sirona' 'Xerox Corp.' 'Xylem Inc.' 'Yum! Brands Inc' 'Zimmer Biomet Holdings' 'Zions
Bancorp' 'Zoetis']
```

K-Means Cluster 0, 1, 3

```
In cluster 0, the following companies are present:
['Apache Corporation' 'Anadarko Petroleum Corp' 'Baker Hughes Inc' 'Chesapeake Energy' 'Cabot Oil & Gas'
'Concho Resources' 'Devon Energy Corp.' 'EOG Resources' 'EQT Corporation' 'Freeport-McMoran Cp & Gld'
'Hess Corporation' 'Hewlett Packard Enterprise' 'Kinder Morgan' 'Marathon Oil Corp.' 'Murphy Oil' 'Noble
Energy Inc' 'Netflix Inc.' 'Newfield Exploration Co' 'National Oilwell Varco Inc.' 'ONEOK' 'Occidental
Petroleum' 'Quanta Services Inc.' 'Range Resources Corp.' 'Southwestern Energy' 'Teradata Corp.'
'Williams Cos.' 'Cimarex Energy']

In cluster 1, the following companies are present:
['Citigroup Inc.' 'Ford Motor' 'Gilead Sciences' 'Intel Corp.' 'JPMorgan Chase & Co.' 'Coca Cola Company'
'Pfizer Inc.' 'AT&T Inc' 'Verizon Communications' 'Wells Fargo' 'Exxon Mobil Corp.']

In cluster 3, the following companies are present:
['Analog Devices, Inc.' 'Alliance Data Systems' 'Alexion Pharmaceuticals' 'Amgen Inc' 'Amazon.com Inc'
'Bank of America Corp' 'BIOGEN IDEC Inc.' 'Celgene Corp.' 'Chipotle Mexican Grill' 'Equinix' 'Edwards
Lifesciences' 'Facebook' 'First Solar Inc' 'Frontier Communications' 'Halliburton Co.' 'Intuitive
Surgical Inc.' "McDonald's Corp." 'Monster Beverage' 'Priceline.com Inc' 'Regeneron' 'TripAdvisor'
 'Vertex Pharmaceuticals Inc' 'Waters Corporation' 'Wynn Resorts Ltd' 'Yahoo Inc.']
```

Hierarchical Clustering Technique - computing cophenetic correlation

```
# list of distance metrics
distance_metrics = ['euclidean','chebyshev','mahalanobis', 'cityblock'] ## Complete the code to add distance metrics

# list of linkage methods
linkage_methods = ['single','complete','average', 'weighted'] ## Complete the code to add linkages
```

```
Cophenetic correlation for Euclidean distance and single linkage is 0.9232271494002922.
Cophenetic correlation for Euclidean distance and complete linkage is 0.7873280186580672.
Cophenetic correlation for Euclidean distance and average linkage is 0.9422540609560814.
Cophenetic correlation for Euclidean distance and weighted linkage is 0.8693784298129404.
Cophenetic correlation for Chebyshev distance and single linkage is 0.9062538164750717.
Cophenetic correlation for Chebyshev distance and complete linkage is 0.598891419111242.
Cophenetic correlation for Chebyshev distance and average linkage is 0.9338265528030499.
Cophenetic correlation for Chebyshev distance and weighted linkage is 0.9127355892367.
Cophenetic correlation for Mahalanobis distance and single linkage is 0.925919553052459.
Cophenetic correlation for Mahalanobis distance and complete linkage is 0.7925307202850002.
Cophenetic correlation for Mahalanobis distance and average linkage is 0.9247324030159736.
Cophenetic correlation for Mahalanobis distance and weighted linkage is 0.8708317490180428.
Cophenetic correlation for Cityblock distance and single linkage is 0.9334186366528574.
Cophenetic correlation for Cityblock distance and complete linkage is 0.7375328863205818.
Cophenetic correlation for Cityblock distance and average linkage is 0.9302145048594667.
Cophenetic correlation for Cityblock distance and weighted linkage is 0.731045513520281.
****************************************************************************************************
Highest cophenetic correlation is 0.9422540609560814, which is obtained with Euclidean distance and average linkage.
```

Hierarchical Clustering Technique - Euclidean distance only

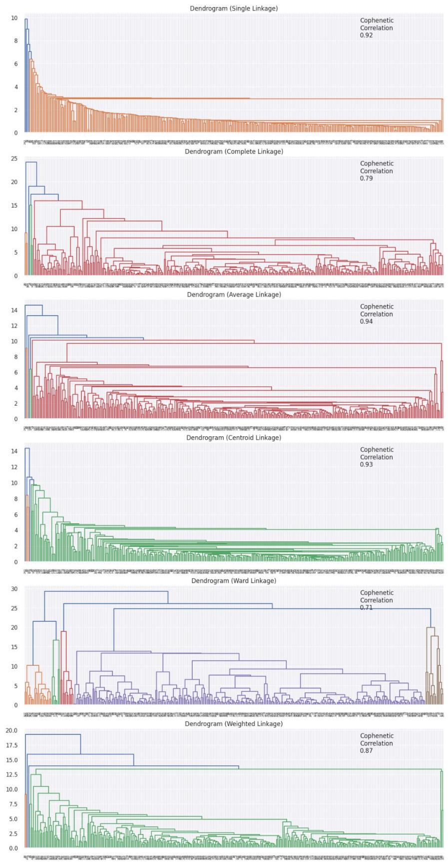
```
# list of linkage methods
linkage_methods = ['single','complete','average', 'weighted', 'ward','centroid'] ## Complete the code to
add linkages
```

```
Cophenetic correlation for single linkage is 0.9232271494002922.
Cophenetic correlation for complete linkage is 0.7873280186580672.
Cophenetic correlation for average linkage is 0.9422540609560814.
Cophenetic correlation for weighted linkage is 0.8693784298129404.
Cophenetic correlation for ward linkage is 0.7101180299865353.
Cophenetic correlation for centroid linkage is 0.9314012446828154.
****************************************************************************************************
Highest cophenetic correlation is 0.9422540609560814, which is obtained with average linkage.
```

- Checking Dendrograms for the following linkage methods using Euclidean Distance only:

```
# list of linkage methods
linkage_methods = ["single", "complete", "average", "centroid", "ward", "weighted"] ## Complete the code
to add linkages
```

Overview Euclidean Distance Dendrograms w/ various linkage methods



Single Linkage Dendrogram. Cophenetic Correlation 0.92

Complete Linkage Dendrogram. Cophenetic Correlation 0.79

Average Linkage Dendrogram. Cophenetic Correlation 0.94

Centroid Linkage Dendrogram. Cophenetic Correlation 0.93

Ward Linkage Dendrogram. Cophenetic Correlation 0.71

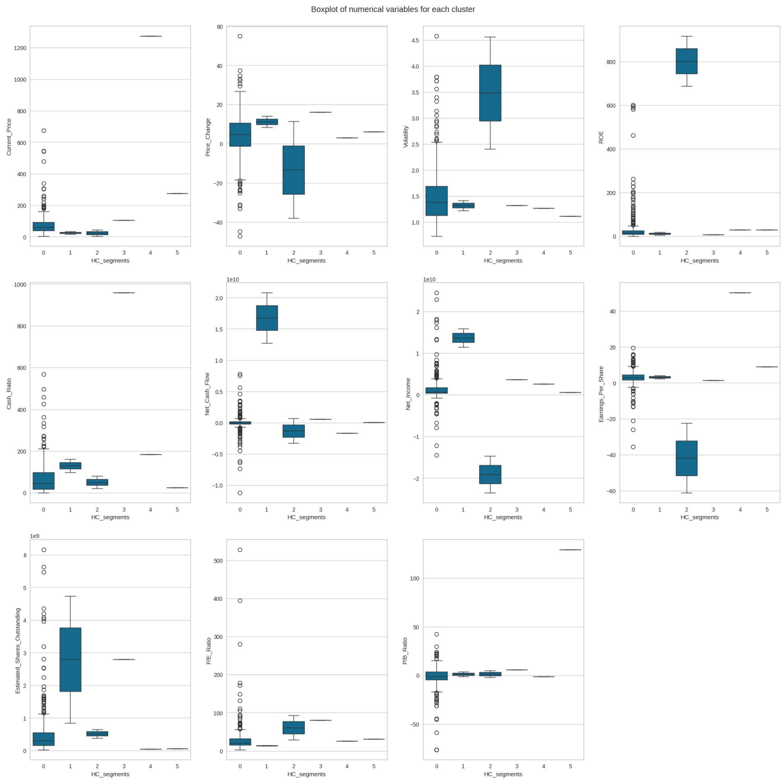
Weighted Linkage Dendrogram. Cophenetic Correlation 0.87

Hierarchical Clustering profiling

```
hc_cluster_profile = df2.groupby("HC_segments")[num_col].mean()
## Complete the code to groupby the cluster labels

hc_cluster_profile["count_in_each_segment"] = (
   df2.groupby("HC_segments")["Security"].count().values
## Complete the code to groupby the cluster labels)

hc_cluster_profile.style.highlight_max(color="lightgreen", axis=0)

## Complete the code to print the companies in each cluster
for cl in df2["HC_segments"].unique():
   print("In cluster {}, the following companies are present:".format(cl))
   print(df2[df2["HC_segments"] == cl]["Security"].unique())
   print()

df2.groupby(["HC_segments", "GICS_Sector"])['Security'].count()
```

```
HC_segments  GICS_Sector
0            Consumer Discretionary         39
             Consumer Staples               19
             Energy                         28
             Financials                     48
             Health Care                    40
             Industrials                    53
             Information Technology         30
             Materials                      20
             Real Estate                    27
             Telecommunications Services     5
             Utilities                      24
1            Financials                      1
             Information Technology          1
2            Energy                          2
3            Information Technology          1
4            Consumer Discretionary          1
5            Information Technology          1
Name: Security, dtype: int64
```

Hierarchical Clustering profiling



Observations:

The 6 clusters compared here are extremely unevenly distributed in terms of datapoint observations.

333 out of 340 companies are grouped together in Cluster 0, and as a result the observations made for that cluster are too general.

On the other hand, the observations for the clusters 1 through 5 are too specfic as they pertain to one or at the most 2 companies only.

K-Means vs Hierarchical Clustering

- It took more preparation time to determine the optimal number of clusters using the K-Means Clustering technique. Several methods available all indicated a similar optimal number of 4.
- The Hierarchical Clustering method indicated 6 as the optimal number of clusters.
- Both algorithms calculated one massive cluster containing a large majority of companies. This was more extreme however in the case of Hierarchical Clustering (333 out of 340 companies vs. 277 out of 340 companies using K-Means), rendering valid observations using Hierarchical Clustering useless.

***Note**: You can use more than one slide if needed*

Slide Header

- Please add any other pointers (if needed)

Happy Learning !